8–EFFECT OF FREE LIQUIDS ON STABILITY

8.1 BASIC CONCEPTS

Consider a rectangular tank of length l and breadth b sitting in a ship of length L and breadth B as shown in Figure 8.1(a). Let C_{LT} be the centreline of tank and C_L that for the ship. The tank contains liquid of density r_L. When the vessel is upright the surface of the liquid will assume the horizontal posture LS.

When the ship is heeled to angle Θ the tank also heels with it leaving behind the liquid. The liquid therefore continues to assume the horizontal surface but at L_1S_1.

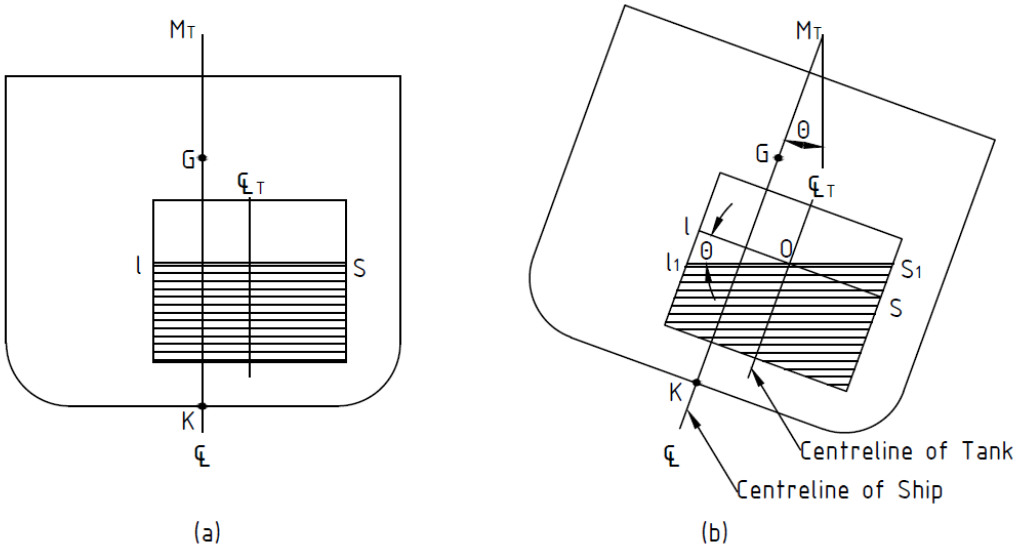


Figure 8.1 – Concept of Free Surface

By superimposing the initial liquid surface LS on the heeled surface L_1S_1 as in Figure 8.1(b), it will appear as if a mass of liquid has rather moved from the port side to the starboard side. This movement will hence cause the C of G to shift accordingly and will affect the stability of the vessel. This is referred to as *FREE SURFACE EFFECT*.

8.2 DETERMINATION OF FREE SURFACE EFFECT (FSE)

Let y represent the half-width of the tank as shown in Figure 8.2(a). For an elemental length dx the mass of liquid assumed to have shifted from the port side:

$$m = \int_0^l ½.y.y\tan\Theta.dx$$

This will cause the C of G of ship to shift transversely as shown in Figure 8.2(b) to give:

$$GG_0 = \frac{\int_0^l \frac{1}{2}.y.y\tan\Theta.dx.2.\frac{2}{3}y.\varrho_L}{\Delta} = \frac{\int_0^l \frac{2}{3}y^3\tan\Theta.dx.\varrho_L}{\Delta}$$

But

$$\int_0^l \frac{2}{3} y^3 dx = i_{CLT}$$

Therefore

$$GG_0 = \frac{\varrho_L i_{CLT} \tan\Theta}{\Delta}$$

Vertically C of G will shift to give:

$$G_0G_1 = \frac{\int_0^l \frac{1}{2}.y.y\tan\Theta.dx.2.\frac{1}{3}y\tan\Theta.\varrho_L}{\Delta} = \frac{\int_0^l \frac{1}{2}.\frac{2}{3}.y^3.\tan^2\Theta.dx.\varrho_L}{\Delta}$$

$$G_0G_1 = \frac{\frac{1}{2}\varrho_L i_{CLT} \tan^2\Theta}{\Delta}$$

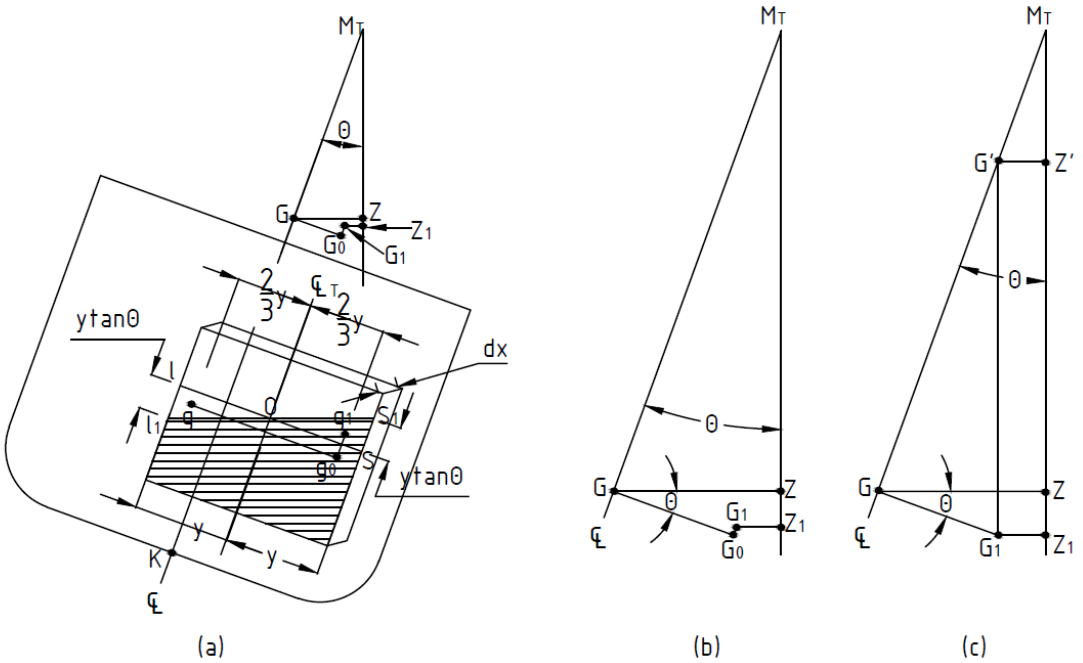


Figure 8.2 – Calculation of FSE

Now for all practical purposes ½tan$^2\Theta$ approximates to zero, therefore G_0G_1 equals zero and G_0 merges with G_1 to reduce GG_0G_1 to GG_1 as shown in Figure 8.2(c). This is to say:

$$GG_1 = GG_0 = \frac{\varrho_L i_{CLT} \tan\Theta}{\Delta}$$

Since displacement is now effectively acting through G_1 then the righting lever GZ has reduced to G_1Z_1. But G_1Z_1 equals G'Z' therefore the metacentric height can be said to have been reduced by the amount GG' to G'M_T. GG' is the measurement for Free Surface Effect.

From triangle GG'G_1:

$$FSE = GG' = \frac{GG_1}{\tan\Theta} = \frac{\frac{\varrho_L i_{CLT} \tan\Theta}{\Delta}}{\tan\Theta}$$

Therefore

$$FSE = \frac{\varrho_L i_{CLT}}{\Delta}$$

Conclusion: It must be stated that the effect of free surface is neither dependent on the height of liquid in a tank nor the position of the tank within the ship.

8.3 MEASURES FOR REDUCING FREE SURFACE EFFECT

8.3.1 Tank without an internal partition

Consider a rectangular tank of length l and breadth b as shown in Figure 8.3 to be placed with its length longitudinally in a ship. Since the second moment of area for a rectangular surface is $lb^3/12$,

$$FSE = \frac{\varrho_L l b^3}{12\Delta}$$

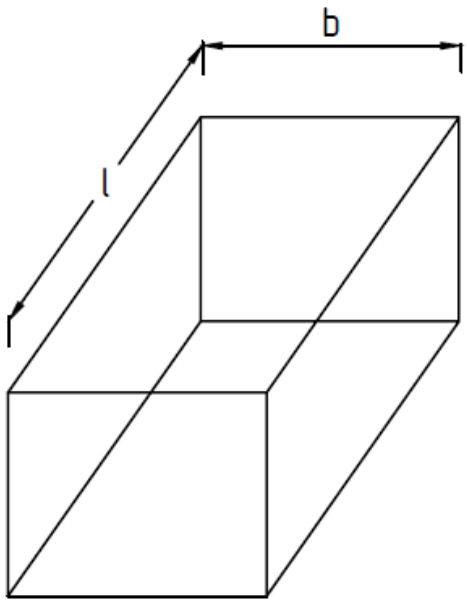


Figure 8.3 – Non-partitioned Tank

8.3.2 Introducing transverse partitions into tanks

In Figure 8.4(a) one watertight partition has been introduced at half length into the original tank of Figure 8.3. This has effectively divided it into two tanks A and B.

$$FSE = FSE_A + FSE_B = \frac{\varrho_L \frac{l}{2} b^3}{12\Delta} + \frac{\varrho_L \frac{l}{2} b^3}{12\Delta}$$

$$\mathrm{FSE} = \frac{\varrho_L l b^3}{12\Delta}$$

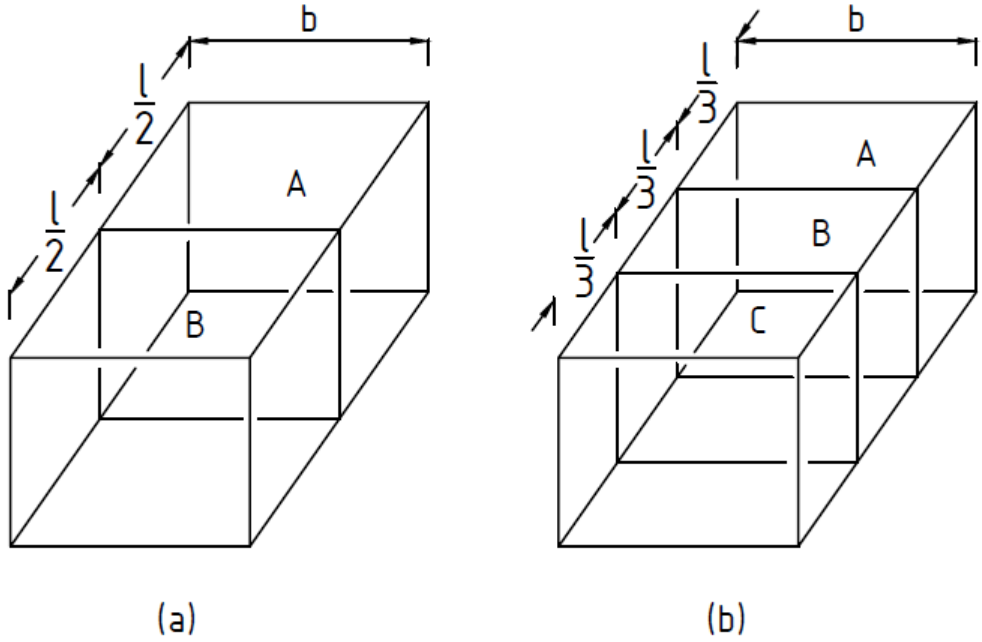


Figure 8.4 – Tanks With Transverse Partitions

Repeating the calculation with two watertight partitions to divide the tank into three A, B, and C as in Figure 8.4(b),

$$\mathrm{FSE} = \mathrm{FSE_A} + \mathrm{FSE_B} + \mathrm{FSE_C} = \frac{\varrho_L \frac{l}{3} b^3}{12\Delta} + \frac{\varrho_L \frac{l}{3} b^3}{12\Delta} + \frac{\varrho_L \frac{l}{3} b^3}{12\Delta}$$

$$\mathrm{FSE} = \frac{\varrho_L l b^3}{12\Delta}$$

Conclusion: Introducing transverse partitions into a tank will not change the effect of free surface.

8.3.2 Introducing longitudinal partitions into tanks

In Figure 8.5(a) one watertight partition has been introduced at half width into the original tank of Figure 8.3. This has effectively divided it into two tanks A and B.

$$\mathrm{FSE} = \mathrm{FSE_A} + \mathrm{FSE_B} = \frac{\varrho_L l \left(\frac{b}{2}\right)^3}{12\Delta} + \frac{\varrho_L l \left(\frac{b}{2}\right)^3}{12\Delta}$$

$$\mathrm{FSE} = \frac{\varrho_L l b^3}{12\Delta} \times \frac{1}{2^2}$$

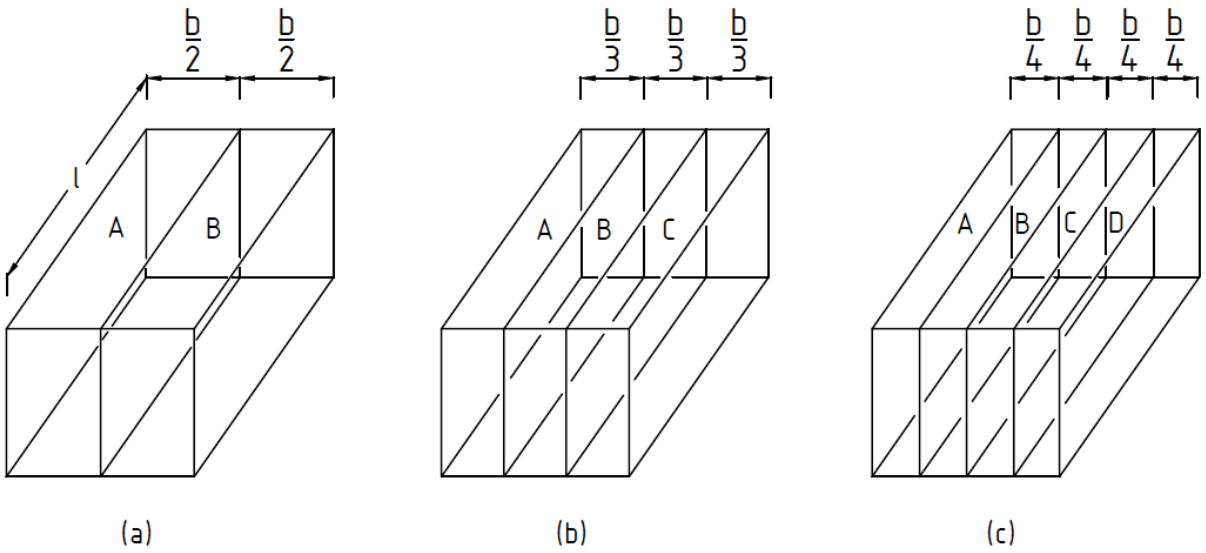


Figure 8.5 – Tanks With Longitudinal Partitions

Once again when the calculation is repeated with two watertight partitions to divide the tank into three A, B, and C as in Figure 8.4(b),

$$\mathrm{FSE} = \mathrm{FSE_A} + \mathrm{FSE_B} + \mathrm{FSE_C} = \frac{\varrho_L l\left(\frac{b}{3}\right)^3}{12\Delta} + \frac{\varrho_L l\left(\frac{b}{3}\right)^3}{12\Delta} + \frac{\varrho_L l\left(\frac{b}{3}\right)^3}{12\Delta}$$

$$\mathrm{FSE} = \frac{\varrho_L l b^3}{12\Delta} \times \frac{1}{3^2}$$

Finally when three partitions are considered to divide the tank in A, B, C and D as in Figure 8.5(c),

$$\mathrm{FSE} = \mathrm{FSE_A} + \mathrm{FSE_B} + \mathrm{FSE_C} + \mathrm{FSE_D} = \frac{\varrho_L l\left(\frac{b}{4}\right)^3}{12\Delta} + \frac{\varrho_L l\left(\frac{b}{4}\right)^3}{12\Delta} + \frac{\varrho_L l\left(\frac{b}{4}\right)^3}{12\Delta} + \frac{\varrho_L l\left(\frac{b}{4}\right)^3}{12\Delta}$$

$$\mathrm{FSE} = \frac{\varrho_L l b^3}{12\Delta} \times \frac{1}{4^2}$$

If (n) is made to represent the number of longitudinal partitions then in general:

$$\mathrm{FSE} = \frac{\varrho_L l b^3}{12\Delta} \times \frac{1}{(n+1)^2}$$

Conclusion: Whenever a number of longitudinal partitions (n) are introduced into a tank, the FSE reduces by an amount equal to the reciprocal of $(n+1)^2$.